KAVIKULGURU INSTITUTE OF TECHNOLOGY AND SCIENCE, RAMTEK

FIRST MID SESSIONAL EXAMINATION (2026-27)

B.TECH. SEVENTH SEMESTER INFORMATION TECHNOLOGY

SUBJECT: ELECTIVE-V NATURAL LANGUAGE PROCESSING (BTIIT703T.1)

Date: 19/08/2026 (AN) Max Marks: 15

Max Time: 1hr.

Note: Assume suitable date wherever necessary.

Q. No		Description	Marks	C.O.
Q.1	a)	Explain the steps involved in text preprocessing and normalization tokenization, stop-word removal, stemming, and lemmatization. Illustrate each step with a suitable example sentence	4	703.1
	b)	Explain the concept of text preprocessing with example.	3	703.1
		OR		
Q.2	a)	Define Natural Language Processing. Explain its scope and describe any four real-world applications of NLP (e.g., machine translation, sentiment analysis, chatbots, speech recognition).	4	703.1
	b)	Define Linguistics. Explain the concept in detail.	3	703.1
Q.3	a)	Explain test categorization of test in detail.	4	703.2
	b)	Explain various n-gram model in detail.	4	703.2
		OR		
Q.4	a)	Explain the Naive Bayes algorithm for text classification. Illustrate with document is classified into a category using Naive Bayes algorithm. Take appropriate working example to demonstrate it.	4	703.2
	b)	Compare Decision Tree and Support Vector Machine (SVM) classifiers for text classification in terms of working principle, advantages, disadvantages, and suitability for high-dimensional text data	4	703.2

PAPER SET BY

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